Lesson plan · for the teacher

One ID per pupil: joining sheets

Data engineering · Level 2 · Lesson 4

Time
40 min

Materials
Class-list cards (a number and a made-up name) and mark cards (a number and a mark); Board and chalk; Optional: a spreadsheet app with XLOOKUP or VLOOKUP, and Python

Learning goals

- Give every pupil one ID.
- Join two sheets by ID, not by name.
- Find and question the rows with no match.

Activity: Human join

1. Warm-up (5 min): Show a short mark list with two pupils of the same name and one spelling difference. Ask pupils to match the marks to the class list by name, and see where it goes wrong.
2. Teach (8 min): Explain a key, a pupil ID given once and never reused, and how a lookup finds the matching row and brings back the name.
3. Join (12 min): Hand out the class-list and mark cards. Pupils find their partner by number and stand together; unmatched pupils stand to one side.
4. Question (10 min): List the unmatched cards and the pupil with two marks on the board. For each, the class decides what it might mean and who to ask.
5. Wrap-up (5 min): If you have a device, show the same join with XLOOKUP or a Python dictionary. Then give the exit ticket.

Prepare the cards before the lesson: one pupil with no mark, one mark with an unknown number and one pupil with two marks. Use made-up names, and keep the number cards so you can reuse them.

Discussion

- Why do names make poor keys?
- What could a mark with no pupil mean?
- Why did one pupil appear twice?

Exit ticket

- Write what a key is in one sentence, and one reason a row might have no match.

Try it in class tomorrow
Play “Human join”: half the class holds class-list cards (a number and a name), the other half holds mark cards (a number). Pupils find their partner by number. Who’s left without a partner, and why?

Worksheet · for pupils

One ID per pupil: joining sheets

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Name _____ Class _____ Date _____

1 **After a lookup, one mark has no matching pupil. What does that tell you?**

- ◯ Delete it: it's rubbish
- ◯ An ID was typed wrong, or a pupil is missing from the list
- ◯ The lookup formula is broken

2 **Two pupils may share the same pupil ID if they are in different classes.**

- ◯ True
- ◯ False

3 **What does this program print?**

```
names = {1: "Amina", 2: "Tunde", 3: "Grace"}
print(names.get(2, "?"), names.get(7, "?"))
```

- ◯ Tunde ?
- ◯ Tunde Grace
- ◯ An error: there is no pupil 7

4 **pupils.csv is the class list and test.csv holds the test marks by pupil ID. Fix the program so it prints each mark with its pupil's name, and ? when the ID isn't on the list, instead of stopping with an error.**

```
import csv
names = {}
with open("pupils.csv") as f:
    for row in csv.DictReader(f):
        names[row["id"]] = row["name"]
with open("test.csv") as f:
    for row in csv.DictReader(f):
        print(names[row["pupil_id"]], row["mark"])
```

Write your program:

5 **Put the lines in order (and indent them): the program prints every pupil on the class list who has no mark.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

- [ ] `for pupil_id in pupils:`
- [ ] `if pupil_id in marked:`
- [ ] `print(pupil_id, "has no mark")`
- [ ] `for pupil_id, mark in marks:`
- [ ] `if pupil_id not in marked:`
- [ ] `marked = set()`
- [ ] `marked.add(pupil_id)`

✎ **Does your school give every pupil one ID that's used on every sheet? If not, which two sheets would you join first, and what would you learn?**